

MD AL EMRAN HOSSAIN

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CAREER OBJECTIVE

Machine Learning Engineer with **4 years** of industry experience specializing in developing and deploying advanced AI systems. I have expertise in **Generative AI**, **Document AI**, and **Applied ML**, with hands-on work across **NLP**, **Computer Vision**, and **Automation**. Skilled in Python and modern AI frameworks, I design scalable, production-grade solutions integrating LLMs, RAG pipelines, and OCR. My work focuses on delivering enterprise-ready AI applications while continuously advancing my technical and research capabilities in the evolving ML landscape.

Future Research Interests: Multimodal ML, DocumentAI, Generative AI, Agentic AI, AI for Health & Education.

EDUCATION

- **BRAC University**, Bangladesh Jan 2026 – Present
MSc in Computer Science and Engineering
MSc Thesis: Multimodal AI (NLP, Computer Vision, and Agentic AI).
Running Courses: Advance Reinforcement Learning & Big Data Analysis
Completed Courses: Advance NLP & Deep Learning for Computer Vision [So far **CGPA: 4.00/4.00**]
 - **United International University (UIU)**, Bangladesh Feb 2018 – Mar 2022
BSc in Computer Science and Engineering **CGPA : 3.90 / 4.00**
Scholastic Award: **Summa Cum Laude**
Courses: Artificial Intelligence, Machine Learning, Pattern Recognition, Bioinformatics, Probability & Statistics
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PROFESSIONAL EXPERIENCE

Jan 2025 – Present | Senior ML Engineer & ML Team Lead

Spectrum Software & Consulting (Pvt.) Ltd.

- Leading a 6+ member ML team under CTO supervision.
- Drive AI/ML initiatives across enterprise projects in document intelligence, RPA, and GenAI.
- Oversee research, model development, and integration across banking, RMG, and logistics domains.
- Architect scalable solutions using LLMs, VLMs, OCR, and RAG pipelines for real-world problems.
- Fine-tune models for enterprise-specific optimization.
- Lead technical strategy for AI-based products like e-KYC, XtractAI, and SpectraTrade.
- Align roadmap with business goals and GenAI advancements.
- Introduced internal tools (e.g., Semi-Auto Annotator, QA Automation) to streamline development cycles.

Jul 2022 – Dec 2024 | Machine Learning Engineer

Spectrum Software & Consulting (Pvt.) Ltd.

- Deployed ML solutions in secure, regulated environments for finance and manufacturing.
- Built modular pipelines for document classification, key-value extraction, and OCR tasks.
- Fine-tuned models (LayoutLMv2, Donut, YOLO, LLaMA) for trade finance and compliance use cases.
- Integrated ML via APIs and automation tools for seamless workflows.
- Developed datasets and annotation protocols for various document types.
- Contributed to tools for model monitoring, retraining, and human-in-the-loop feedback.
- Authored documentation for model behavior and deployment strategies.

Feb 2022 – Jun 2022 | Research Assistant

United International University (UIU)

- Conducted research on cyberbullying detection across social media platforms under the supervision of Dr. Saddam Hossain Mukta.
- Contributed to a funded project by the Institute for Advanced Research (IAR), involving data collection, preprocessing, and model experimentation.

Nov 2021 – Feb 2022 | Undergraduate Teaching Assistant

United International University (UIU)

- Supported *Pattern Recognition* and *Database Management* labs by guiding student projects and assisting with coursework.
- Evaluated assignments, quizzes, and lab reports, ensuring timely feedback and academic support.

RESEARCH & PUBLICATIONS

- **Undergraduate Thesis** [2024]
Deep Transfer Learning Based Peer Review Aggregation and Meta-Review Generation for Scientific Articles
arXiv preprint [arXiv:2410.04202](https://arxiv.org/abs/2410.04202)
DOI: [10.48550/arXiv.2410.04202](https://doi.org/10.48550/arXiv.2410.04202)
- **Journal Article** [2023]
A Review on Deep-Learning-Based Cyberbullying Detection
Future Internet, 15(5), 179, 2023
DOI: [10.3390/fi15050179](https://doi.org/10.3390/fi15050179)
- **Conference Paper** [2022]
Adaptive Tabu Dropout for Regularization of Deep Neural Networks
International Conference on Neural Information Processing (ICONIP), pp. 355–366, Springer, 2022
DOI: [10.1007/978-3-031-30105-6_30](https://doi.org/10.1007/978-3-031-30105-6_30)

PROJECTS

Industry Projects & Experimental Work

XtractAI – SSCL's Document AI Platform (**ML Lead**) [2025-Present]
(Ongoing development: continuously integrating new Document AI capabilities)

- Built a live demo platform to showcase SSCL's in-house capabilities in Document AI, enabling real-time document processing and intelligent information extraction.
- Led the development and deployment of the machine learning components, including model integration and backend processing pipelines.
- Enabled automated extraction of key data from diverse document types (e.g., PDFs, forms, images), demonstrating practical AI solutions for enterprise use cases.
- Provided live product access for evaluation with *1,000 free credits* upon Gmail login for testing and validation purposes.

e-KYC – Electronic Know Your Customer System (Deployed in [TAP App](#)) [2025-Present]
[System Architect | Engineering Manager | Lead Developer]

- Led end-to-end product strategy, architecture, and developed of an AI-powered e-KYC system enabling secure, automated digital onboarding.
- API integration with the Bangladesh Election Commission for real-time NID verification and facial matching.
- Built an advanced image processing + OCR pipeline (dewarping, blur/light detection, angle correction, enhancement) improving extraction reliability.
- Fine-tuned Vision-Language Models (VLMs) achieving 97.48%+ (English) and 91.23% (Bangla) NID text extraction accuracy using synthetic Smart & Paper ID datasets.
- Implemented instruction-based video liveness detection and live-to-NID face matching to prevent identity fraud.
- Designed an administrative [e-KYC portal](#) with analytics, monitoring, and role-based access control.

Import Document Scrutiny (Project Lead | System Architect) 2025
(Achieved 82% reduction in cost and manual effort)

- Automated cross-validation of MT700 Letter of Credit (LC) clauses against submitted import documents.
- Validated the structure and completeness of import documents in accordance with UCP 600 and ISBP 821 rulebooks using a RAG-based framework.
- Converted unstructured LC messages into executable business logic and verified them against automatically extracted data from classified trade documents.
- Implemented a multi-modal AI approach using a fine-tuned GPT-OSS-20B model integrated with DotOCR for accurate text and layout understanding.

SpectraTrade – Financial Document Classification & Extraction (**ML Lead**) [2023-2025]

- Achieved high-accuracy financial document classification and information extraction using GPT, LLaMA (3.5 Vision finetuned), Qwen, Gemma, and Donut models.
- Optimized processing speed to 3–5 seconds per page, improving efficiency for large-scale banking operations.
- Deployed in **PrimeBanijo**, the country's first AI-powered digital LC management platform, enabling faster, more accurate, and seamless trade operations.

DockStream – RMG Sector Automation (**Project Manager & ML Lead**) [2024-2025]

- Led automation of document classification, data entry, form filling, and information submission across multiple websites using RPA technologies.
- Designed adaptive pipelines for extracting data from structured and unstructured documents, enabling end-to-end process automation.
- Achieved an 86% reduction in manual effort, human error, and associated operational costs.

SQA Automation [2025]

- Developed an internal productivity tool for QA operations, adapting to dynamic UI and operational changes based on client requirements.
- Minimized manual test case writing after each change, significantly reducing QA effort and accelerating release cycles.
- Developed as an internal productivity tool.

Robotic Process Automation (RPA) & GenAI for Financial Document Processing [2023-2025]

- Led OCR pipeline development supporting **24+ financial document** types with multi-format JSON outputs.
- Extracted key data from financial documents using Faster R-CNN, Donut, YOLOv5/v8, LayoutLMv2, and fine-tuned Donut and LLaMA models for improved accuracy.
- Developed financial document classification and table extraction (detection & segmentation) leveraging LayoutLMv2 and related techniques.

Semi-Auto Annotator & DocTrainer [2025]

- Developed a tool for automated PDF annotation with human-in-the-loop feedback (RLHF).
- Enabled iterative model retraining for continuous accuracy improvement.

MLOps Experiences

- eKYC Platform** – SaaS-Based Identity Verification Solution [2026]
- Led end-to-end MLOps deployment of a SaaS-based eKYC platform, deploying NID OCR, Liveness Detection, and EC Verification services through secure APIs with scalable inference infrastructure, load balancing, and NGINX-based hosting.
- SpectraTrade** – Secured Banking Deployment (*Leading Private Bank in Bangladesh*) [2023-2024]
- Led end-to-end MLOps deployment of ML modules (LC Management, Shipping Guarantee) on RHEL-secured on-premises infrastructure.
 - Established the initial ML runtime environment, managed model lifecycle across UAT and Production, and integrated ML services via secure, API-based interfaces.
- DockStream** - RMG Sector Automation (Enterprise ML Deployment) [2024–2025]
- Led end-to-end MLOps deployment of ML-driven document intelligence and RPA automation pipelines for large-scale RMG operations.
 - Operationalized document classification, structured/unstructured data extraction, and form automation models, integrating ML services with RPA workflows via production-grade APIs, resulting in an 86% reduction in manual effort and operational errors.

Academic Research & Projects

- ZeroDepthFAS** – Passive Human Face Liveness Detection [MSc Course Work & Ongoing Research] [2026]
- Built a passive face anti-spoofing system fusing Vision Transformer (Large) with EfficientNet-B0 for real-time e-KYC liveness detection.
 - Achieved AUC 1.000 in-distribution and HTER 1.48% on zero-shot out-of-distribution replay attacks; deployed as a REST API with sub-second GPU inference.
- HealthCareGPT** – LLM + RAG for Disease Prediction [Course Work] [2024]
- Designed a healthcare assistant for disease prediction & personalized recommendations using LLMs and Retrieval-Augmented Generation (RAG).
 - Demonstrated higher accuracy than traditional ML methods.
- RLHF-Enhanced Document Classification (LayoutLM)** [Course Work] [2024]
- Integrated Reinforcement Learning with Human Feedback (RLHF) to improve LayoutLM-based financial document classification.
- UIU EMAS** – Event Management & Archiving System [Team Project] [2022]
- Web-based system for managing the CSE Project Show & Club Forum Fest.
 - Designed as part of a Software Engineering course project.

TECHNICAL EXPERTISE

- **Programming Languages:** Python*, C/C++, Java, PHP, Assembly
- **Machine Learning & Deep Learning**
 - **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face (Transformers, Datasets, Accelerate)
 - **ML & DL Experience:** Linear/Logistic Regression, Decision Trees, Random Forests, SVM, KNN, K-Means, Naive Bayes, PCA, XGBoost, CNN, RNN, LSTM, Transformers, RL, RLHF, BERT, Donut, YoLo, etc.
- **Generative AI Ecosystem**
 - **Fine-Tuning LLMs/VLMs:** LLaMA, Qwen, Mistral, Granite, DeepSeek, Gemma, GPT-OSS
 - **API-Based LLMs:** OpenAI, Anthropic (Claude), Google Gemini, Alibaba Dashscope, Together AI
 - **OCR Tools:** DeepSeek-OCR, Chandra, Surya, LightOn, Paddle, EasyOCR, Tesseract, TrOCR, olmOCR, etc.
 - **Frameworks & Tools:** LangChain, Hugging Face Transformers, Prompt Engineering, Vision Transformers, MCP, MLOps Workflows
 - **RAG & Knowledge Systems:** RAG Pipelines (including Graph RAG with Neo4j), Vector Databases (Pinecone, FAISS), Adaptive Chunking Strategies
- **Deployment, Inference & Application Frameworks**
 - **Inference Engines:** vLLM, Ollama, Together AI, TensorRT
 - **Web & API Frameworks:** Flask, FastAPI, Gradio, Django
 - **Cloud:** AWS (foundational experience), S3, API Development & Hosting
 - **Workflow Automation & Orchestration:** n8n, CI/CD Workflows, Jira
- **Testing, Scripting & Agents:** Playwright, Selenium, PyTest, Computer-Use Agents (CUAs)
- **Databases & Storage:** MySQL, PostgreSQL, SQLite
- **DevOps & Version Control:** Git, GitHub, GitLab, Docker, Nginx
- **Operating Systems:** Linux (Ubuntu, RHEL/CentOS), Windows
- **GPU Setup & Used:** Nvidia A4000, A4500 Ada, A5000, A6000, DGX Spark (ARM64)
- **Data Engineering & API Development:** Data Preprocessing, Feature Engineering, API Development, Integration of AI models into production pipelines

ACHIEVEMENTS

- Scholastic Award **Summa Cum Laude** at 7th Convocation of UIU. [2024]
 - Out of 12 trimesters, 8x100% and 3x50% Scholarship for outstanding results at UIU. [2018-2022]
 - **Champion** in UIU CSE Project Show (Fall 2022) for "*UIU EMAS*". [2022]
 - **2nd Runner-Up** in UIU CSE Project Show for "*Pick-a-Book*". [2021]
 - **General Board Scholarship** in *HSC* exams. [2016]
 - **Talent Pool Scholarship** (9th district rank) in *SSC*. [2014]
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EXTRACURRICULAR

- **Instructor** at Programmer Doctor's Chamber, addressing programming challenges. [2021-2022]
 - **Programming Wing** Executive and **Media & Communication** Executive at *UIU Computer Club*. [2018-2022]
 - **Founding Member** of *UIU Mars Rover Team*. [2021]
 - **General Member** of *UIU Robotics Club*. [2018-2022]
 - **Volunteer** at *ICAICT 2020* and various national/international events. [2022]
 - Former Member of *Bangladesh Biggan Porishod*. [2016-2018]
 - Former **Scout Group Leader** at *Shahebrampur High School*. [2012-2014]
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REFERENCES

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| <ul style="list-style-type: none">• Dr. Faizul Bari (Faiz), PhD
Chief Technology Officer (CTO),
<i>Spectrum Software & Consulting Ltd.</i>
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